Finding the Inverse of a Function

Algebra Worksheet · Grade 9–11

Name: ________________________________ Date: ________________

Learning Objectives

- Understand inverse function notation and the concept of swapping domain and range
- Find the inverse of a function algebraically by interchanging x and y and solving for y
- Find the inverse of radical and other non-linear functions using algebraic manipulation

Problems

1. A function is given in table form. Write the inverse function as a new table by interchanging the x and y values.

2. Find the inverse of the function below.

$$f(x) = x + 5$$

3. Find the inverse of the linear function below.

$$f(x) = 2x + 7$$

4. Find the inverse of the function below.

$$f(x) = 3x - 4$$

5. Find the inverse of the radical function below.

$$f(x) = \sqrt{x + 4}$$

6. Find the inverse of the radical function below.

$$f(x) = \sqrt{2x - 3}$$

7. Find the inverse of the function below.

$$f(x) = \frac{x+1}{3}$$

8. Find the inverse of the function below.

$$f(x) = \frac{2x+5}{4}$$

9. Find the inverse of the cubic function below.

$$f(x) = x^3 - 2$$

10. Find the inverse of the function below, then evaluate the inverse at x equals 3.

$$f(x) = \frac{5x-1}{2}$$

Finding the Inverse of a Function — Answer Key

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Answer Key

1. Answer: Inverse table: x = 1, 3, 5, 7 and y = -2, 0, 2, 4

- To find the inverse from a table, swap every x and y value.
- The inverse table becomes: x: 1, 3, 5, 7 and y: -2, 0, 2, 4

2. Answer: $f^{-1}(x) = x - 5$

- Replace f(x) with y: $y = x + 5$
- Interchange x and y: $x = y + 5$
- Solve for y: $y = x - 5$
- Write as inverse: $f^{-1}(x) = x - 5$

3. Answer: $f^{-1}(x) = (x - 7) / 2$

- Replace f(x) with y: $y = 2x + 7$
- Interchange x and y: $x = 2y + 7$
- Subtract 7 from both sides: $x - 7 = 2y$
- Divide both sides by 2: $y = (x - 7) / 2$
- Write as inverse: $f^{-1}(x) = (x - 7) / 2$

4. Answer: $f^{-1}(x) = (x + 4) / 3$

- Replace f(x) with y: $y = 3x - 4$
- Interchange x and y: $x = 3y - 4$
- Add 4 to both sides: $x + 4 = 3y$
- Divide both sides by 3: $y = (x + 4) / 3$
- Write as inverse: $f^{-1}(x) = (x + 4) / 3$

5. Answer: $f^{-1}(x) = x^2 - 4$

- Replace f(x) with y: $y = \text{sqrt}(x + 4)$
- Interchange x and y: $x = \text{sqrt}(y + 4)$
- Square both sides: $x^2 = y + 4$
- Subtract 4 from both sides: $y = x^2 - 4$
- Write as inverse: $f^{-1}(x) = x^2 - 4$

6. Answer: $f^{-1}(x) = (x^2 + 3) / 2$

- Replace f(x) with y: $y = \text{sqrt}(2x - 3)$
- Interchange x and y: $x = \text{sqrt}(2y - 3)$
- Square both sides: $x^2 = 2y - 3$
- Add 3 to both sides: $x^2 + 3 = 2y$
- Divide both sides by 2: $y = (x^2 + 3) / 2$
- Write as inverse: $f^{-1}(x) = (x^2 + 3) / 2$

7. Answer: $f^{-1}(x) = 3x - 1$

- Replace f(x) with y: $y = (x + 1) / 3$
- Interchange x and y: $x = (y + 1) / 3$
- Multiply both sides by 3: $3x = y + 1$
- Subtract 1 from both sides: $y = 3x - 1$
- Write as inverse: $f^{-1}(x) = 3x - 1$

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8. Answer: $f^{-1}(x) = (4x - 5) / 2$

- Replace f(x) with y: $y = (2x + 5) / 4$
- Interchange x and y: $x = (2y + 5) / 4$
- Multiply both sides by 4: $4x = 2y + 5$
- Subtract 5 from both sides: $4x - 5 = 2y$
- Divide both sides by 2: $y = (4x - 5) / 2$
- Write as inverse: $f^{-1}(x) = (4x - 5) / 2$

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9. Answer: $f^{-1}(x)$ = cube root of $(x + 2)$

- Replace f(x) with y: $y = x^3 - 2$
- Interchange x and y: $x = y^3 - 2$
- Add 2 to both sides: $x + 2 = y^3$
- Take the cube root of both sides: $y = (x + 2)^{(1/3)}$
- Write as inverse: $f^{-1}(x) = \sqrt[3]{(x + 2)}$

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10. Answer: $f^{-1}(x) = (2x + 1) / 5$; $f^{-1}(3) = 7/5$

- Replace f(x) with y: $y = (5x - 1) / 2$
- Interchange x and y: $x = (5y - 1) / 2$
- Multiply both sides by 2: $2x = 5y - 1$
- Add 1 to both sides: $2x + 1 = 5y$
- Divide both sides by 5: $y = (2x + 1) / 5$
- Write as inverse: $f^{-1}(x) = (2x + 1) / 5$
- Evaluate at x = 3: $f^{-1}(3) = (2(3) + 1) / 5 = 7/5$
